Catholic Junior College

JC2 Preliminary Examinations

Higher 2

CANDIDATE NAME	
CLASS	**2T**

PHYSICS 9478/02

Paper 2 Structured Questions **27 August 2026**

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name and class in the boxes at the top of this page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

FOR EXAMINER'S USE	
Q1	/ 8
Q2	/ 11
Q3	/ 11
Q4	/ 12
Q5	/ 5
Q6	/ 8
Q7	/ 20
PAPER 2	/ 75

DATA:

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$(\frac{1}{4\pi\varepsilon_0}$	=	8.99×10^{9} m F$^{-1})$
elementary charge	e	=	1.60×10^{-19} C
Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
Boltzmann constant	k	=	1.38×10^{-23} mol^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

FORMULAE:

uniformly accelerated motion	s	=	$u\,t + \frac{1}{2}a\,t^2$
	v^2	=	$u^2 + 2\,a\,s$
work done on / by a gas	W	=	$p\,\Delta V$
pressure	p	=	$\frac{F}{A}$
gravitational potential	ϕ	=	$-\frac{GM}{r}$
temperature	T / K	=	$T / {}^{\circ}C + 273.15$
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	E	=	$\frac{3}{2}kT$
displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t = \pm\omega\sqrt{\left(x_0^{\,2} - x^2\right)}$
electric current	I	=	$nAvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C}$	=	$\frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	C	=	$C_1 + C_2 + \ldots$

energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0\, e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_o r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_o I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_o NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_o nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0\, e^{-\lambda t}$
decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

1 **(a)** Define *gravitational potential at a point*.

……

…… [1]

(b) A satellite of mass m orbits a planet of mass M and radius R_p. The radius of the orbit is R. The satellite and the planet may be considered to be point masses with their masses concentrated at their centres. They may be assumed to be isolated in space.

For the satellite in its orbit, show that

(i) its kinetic energy E_k is given by $E_k = \frac{GMm}{2R}$,

[2]

(ii) the ratio $\frac{\text{kinetic energy of satellite } E_k}{\text{total energy of satellite } E_T}$ is equal to -1.

[2]

(c) The satellite in **(b)** gradually loses energy due to small resistive forces.

Suggest why many such satellites eventually 'burn up' in the Earth's atmosphere.

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……

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…… [3]

[Total: 8]

2 Figure 2.1 shows a Kundt's tube of length 35 cm, containing air and fine sand, having one end open at position **M** and one closed at position **N**. A loudspeaker connected to a sound generator is placed at its open end. When the sound generator and loudspeaker is turned on, four small heaps of sand eventually formed at certain positions in the tube. Adjacent heaps of sand are at distance 10 cm apart as shown in Figure 2.1.

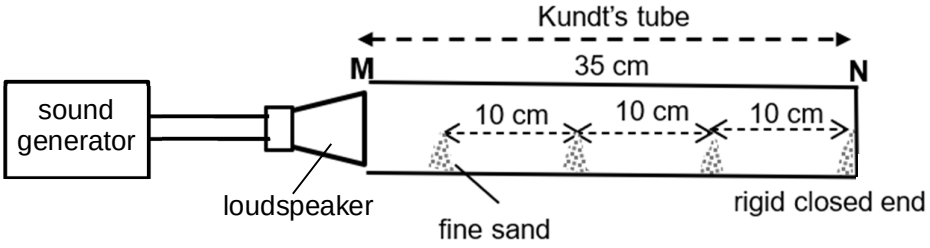


Figure 2.1

(a) Explain how a stationary wave is formed in the tube.

……

……

……

……

……

……

…… [2]

(b) Determine the wavelength of the sound produced by the loudspeaker.

wavelength = …….. cm [1]

(c) (i) The sound wave produced by the loudspeaker in Figure 2.1 has amplitude of A_o.

On Figure 2.2, sketch a graph of the variation of amplitude, in terms of A_o, of the stationary sound wave with distance x along the Kundt's tube, starting from point **M** of the Kundt's tube (x = 0 cm), until its closed end at point **N** (x = 35 cm). You may assume there is no energy loss by the sound wave as it is reflected at the closed end of the tube.

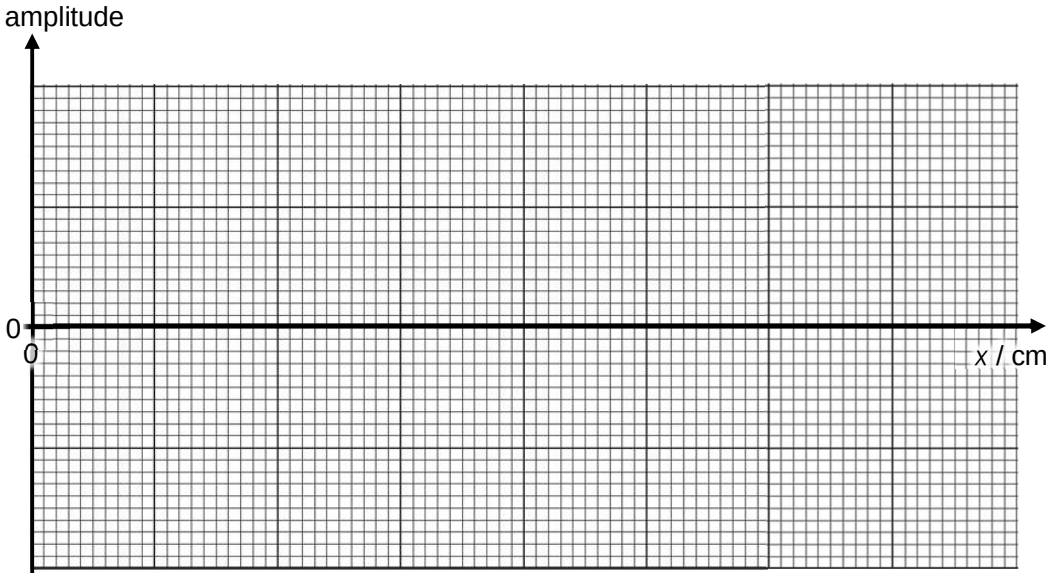


Figure 2.2 [3]

(ii) By describing the movement of air molecules in the stationary sound wave, state and explain where the air pressure varies the most.

……

……

……

……

……

……

……

…… [3]

(iii) The maximum air pressure in the Kundt's tube is P_o.

On Figure 2.3, sketch a graph of air pressure variation, in terms of P_o, against distance x along the Kundt's tube from point **M** (x = 0) to point **N** (x = 35 cm).

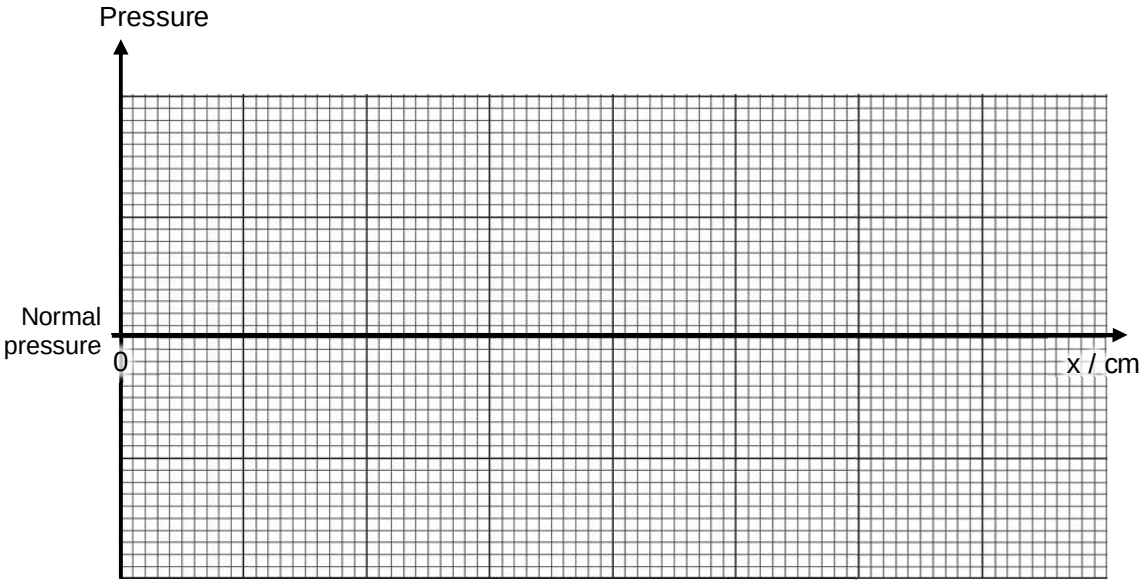


Figure 2.3 [2]

[Total: 11]

3 **(a)** The kinetic theory of gases is based on some simplifying assumptions. For example, the molecules of the gas are assumed to behave as hard elastic identical spheres.

State the basic assumption about the molecules of an ideal gas based on

(i) the nature of their movement,

……..…

……..……... [1]

(ii) their volume.

……..…..…

……..…

……..……... [2]

(b) A cube of volume V contains N molecules of an ideal gas. Each molecule has a component c_x of velocity normal to one side S of the cube, as illustrated in Figure 3.1.

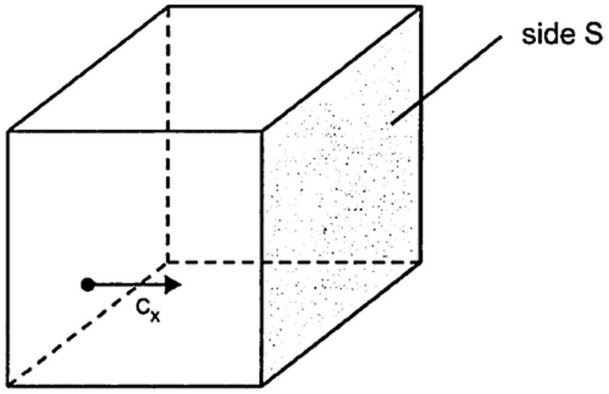


Figure 3.1

The pressure p of the gas due to the component c_x of velocity is given by the expression

$$pV = Nm\langle c_x^2 \rangle$$

where m is the mass of a molecule and $\langle c_x^2 \rangle$ is the mean-square-speed of the molecules in the direction normal to side S of the cube in Figure 3.1.

Show and explain how the above expression leads to the relation

$$pV = \frac{1}{3} Nm\langle c^2 \rangle$$

where $\langle c^2 \rangle$ is the mean-square-speed of the molecules.

[3]

(c) Neon gas behaves as an ideal gas.

A mass of 20.0 g of neon gas occupies a volume of 2.24×10^4 cm^3 at a pressure of 1.01×10^5 Pa and a temperature of 273 K.

Deduce the root-mean-square (r.m.s.) speed of neon atoms at a temperature of

(i) 273 K,

r.m.s. speed = ……..…….…..m s^{-1} [3]

(ii) 27 ˚C.

r.m.s. speed = ……..…….…..m s^{-1} [2]

[Total: 11]

4 **(a)** By reference to energy, explain what is meant by *electromotive force* (e.m.f.).

……...

……...

…… [1]

(b) A 'potato cell' is formed by inserting a copper plate and a zinc plate into a potato. The circuit shown in Figure 4.1 is used in an investigation to determine the e.m.f. and internal resistance of the potato cell.

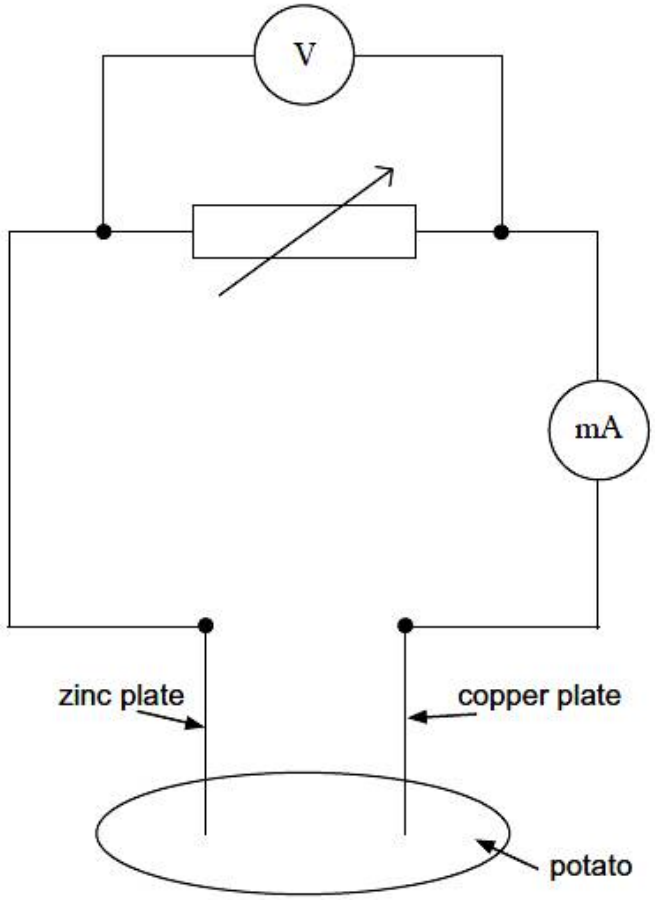


Figure 4.1

The plotted points × on Figure 4.2 show the data for current and voltage that were obtained by varying the resistance of the variable resistor. The line of best fit was then drawn through the plotted points.

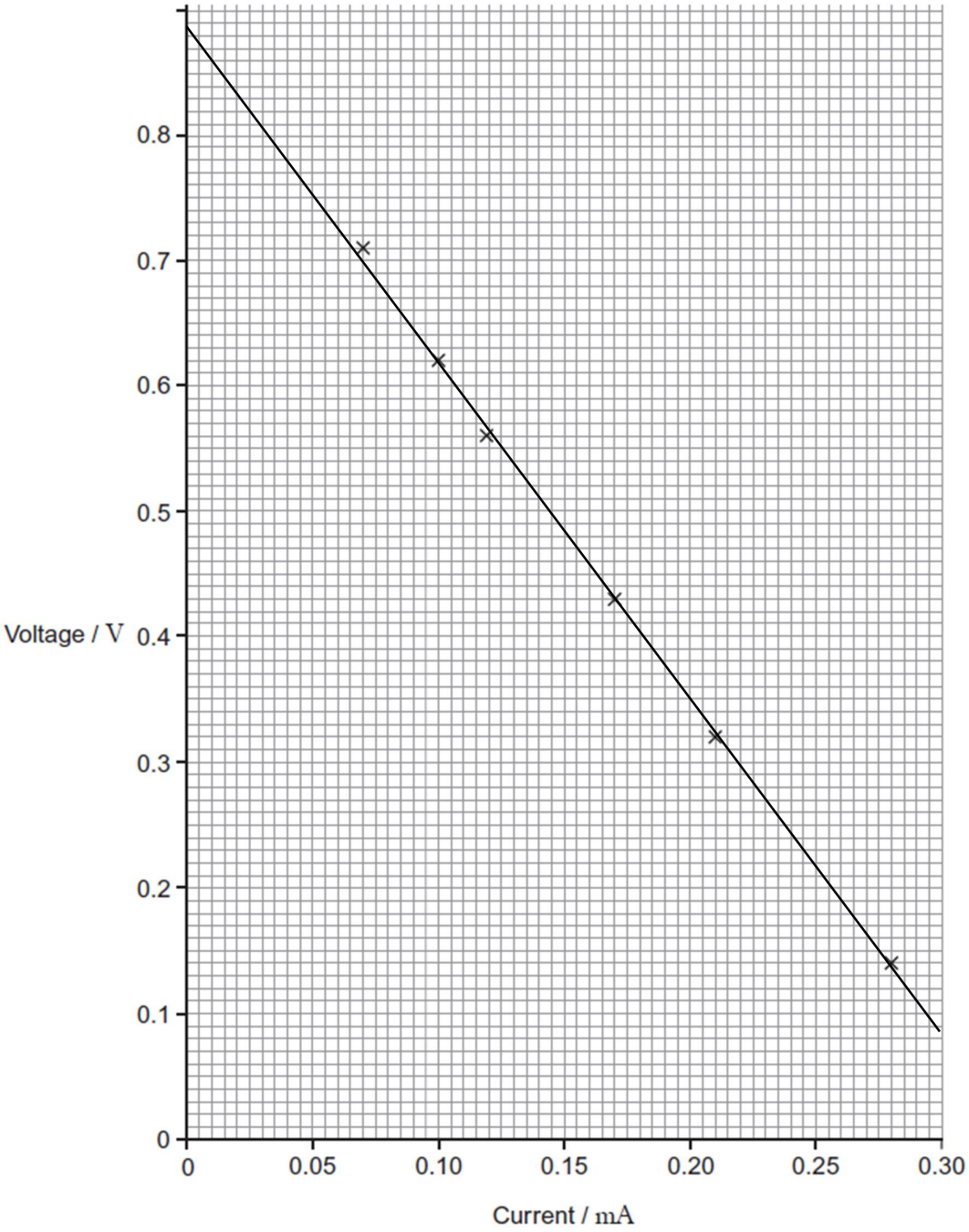


Figure 4.2

(i) State the e.m.f. of the potato cell.

e.m.f. = ……V [1]

(ii) Determine the internal resistance of the potato cell.

internal resistance = ……Ω [3]

(iii) Determine the voltage across the variable resistor when the power delivered to it is maximum. Explain your working.

voltage = ……V [3]

(iv) Explain why the voltages of the plotted points × on Figure 4.2 are always less than the e.m.f. of the potato cell.

……..……..

……..……..

……..……..

……..……..

……..……..

……..……..

……..…… [3]

(v) In the space below, draw a circuit diagram to suggest a modification to the circuit in Figure 4.1 which allows the e.m.f. value of the potato cell to be read directly from the voltmeter.

[1]

[Total: 12]

5 **(a)** A simple a.c. generator which consists of a rectangular coil with 40 turns rotates at 50 revolutions per second in a uniform magnetic field of flux density 50 mT. The coil is 50 cm long and 35 cm wide, as shown in Figure 5.1. An electromotive force (e.m.f.) is induced in the coil.

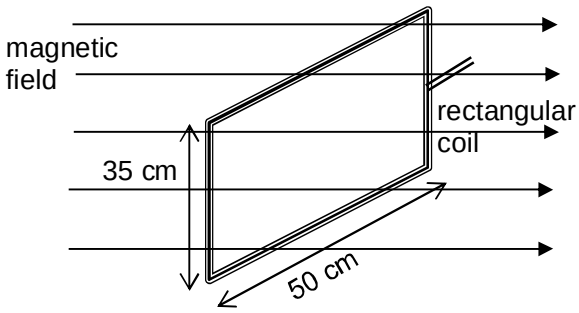


Figure 5.1

The maximum induced e.m.f. E_o in the coil is given by the expression

E_o = maximum magnetic flux linkage × angular speed of the coil

Use this expression to calculate the maximum e.m.f. induced E_o.

E_o = ……V [2]

(b) The output from the generator in part **(a)** is connected to the input of a transformer. The transformer has 40 turns on its primary coil and 10 turns on its secondary coil. The transformer may be considered to be ideal. The secondary coil is connected in series to a resistor of resistance 100 Ω and a diode, as shown in Figure 5.2.

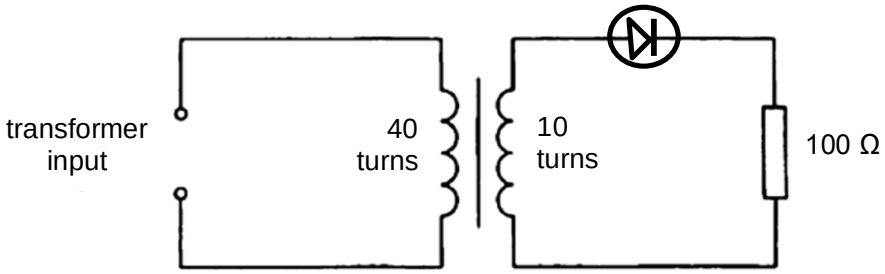


Figure 5.2

Calculate the root-mean-square voltage across the 100 Ω resistor.

root-mean-square voltage = ……V [3]

[Total: 5]

6 An electron is confined within a one-dimensional box of width L.

ψ represents the sinusoidal wavefunction of the electron. The graph of $|\psi|^2$ against distance x from one wall of the box is shown in Figure 6.1.

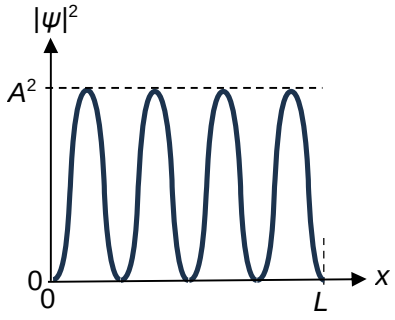


Figure 6.1

(a) **(i)** State what is represented by $|\psi|^2$.

……

…… [1]

(ii) The graph of $|\psi|^2$ against x in Figure 6.1 has an amplitude of A^2.

Derive an expression for A in terms of L.

A = …… [2]

(b) The electron with the $|\psi|^2$ graph shown in Figure 6.1 is in an excited quantum state n in a box where L = 0.10 nm.

(i) State the quantum state n of the trapped electron shown in Figure 6.1.

n = …… [1]

(ii) Calculate the momentum of the electron at this quantum state.

momentum =…… N s [2]

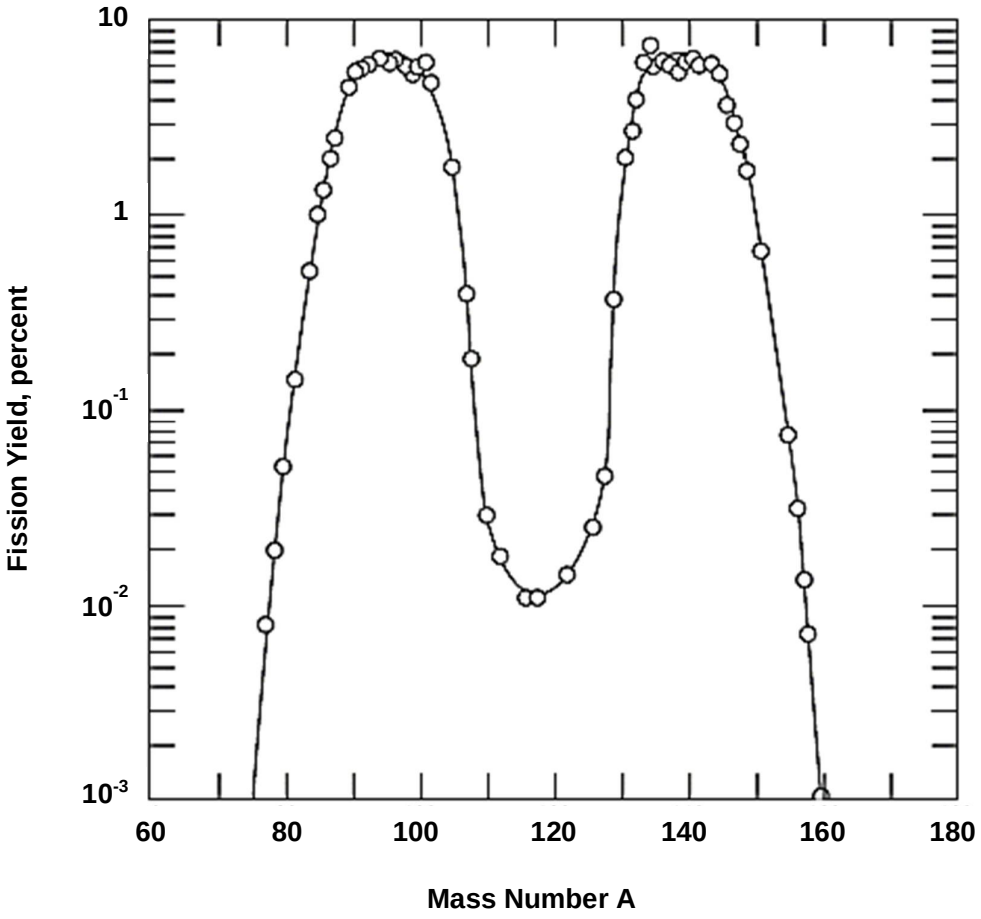
(iii) Calculate the energy, in eV, of the electron at this quantum state.

energy = ……..…… eV [2]

[Total: 8]

7 When a very strong earthquake of magnitude 9.0 hit Japan on 11 March 2011, the accompanying tsunami caused the devastation of the Fukushima Dai-ichi Nuclear Power Plant (Fk-1). A large amount of radioactive material leaked out of Fk-1 as a result, mainly through cracks made by the quake on some reactors, explosions, intentional vents to relieve pressure, and leakage of cooling water that was contaminated with the melted fuel rod debris.

Fk-1 was capable of generating up to 4.7 gigawatts of power through neutron-induced fissions of Uranium-235 (^{235}U) nuclei. Fission resulting from neutron absorption is called *induced* fission. Some nuclides can also undergo *spontaneous* fission, but that is quite rare. Nuclear fission is a decay process in which an unstable nucleus splits into two fragments of comparable mass, although nearly equal mass is unlikely. Over 100 different nuclides, representing more than 20 different elements, have been found among the fission products. Figure 7.1 shows the distribution of mass numbers for fission fragments from the fission of ^{235}U. Fission yield refers to the percentage of a nuclide of a fission fragment produced per fission. Two or three free neutrons usually appear along with the fission fragments.



Source: https://www.quora.com/Why-is-promethium-produced-from-uranium-235

Figure 7.1

The induced fission of ^{235}U may be represented by a nuclear equation of the form

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{a}_{b}\mathrm{P} + {}^{c}_{d}\mathrm{Q} + (2 \text{ or } 3)\,{}^{1}_{0}\mathrm{n} + \text{energy}$$

where P represents the lower mass number fission fragment, and Q represents the higher mass number fission fragment.

One day after the earthquake, elevated levels of Iodine-131, Cesium-134 and Cesium-137, which are common fission fragments of Uranium-235, were detected near the nuclear plant site. Figure 7.2 shows some general information of the three fission fragments.

Isotope	Symbol	Half-life
Iodine-131	$^{131}_{53}\mathrm{I}$	8.0197 days
Cesium-134	$^{134}_{55}\mathrm{Cs}$	2.065 years
Cesium-137	$^{137}_{55}\mathrm{Cs}$	30.17 years

Figure 7.2

(a) Explain the difference between *induced* and *spontaneous* fission.

……...

……...

…… [1]

(b) Suggest why the *percentage fission yield* in Figure 7.1 is shown on a logarithmic scale.

……...

……...

…… [1]

(c) Using Figure 7.1, state the two most common ranges of mass numbers of the fission fragments (P and Q) of ^{235}U.

most common range of mass numbers of P: …… [1]

most common range of mass numbers of Q: …… [1]

(d) **(i)** Uranium-235 undergoes nuclear fission to produce Rubidium-97 and Cesium-137.

Write down the nuclear equation to represent this reaction. Use the chemical symbol Rb to represent Rubidium.

……... [2]

(ii) With reference to Figure 7.1, state the percentage fission yield of Cesium-137 and Rubidium-97.

percentage yield of Cesium-137= ……%

percentage yield of Rubidium-97 = ……% [1]

(iii) Suggest why both the answers in **(d)(ii)** are similar.

……

……

……... [1]

(iv) Determine the **two** possible mass numbers of the other fission fragment produced if one of the fragments has a mass number of 81 in the nuclear fission of Uranium-235.

mass number = ……. or ……..... [2]

(e) Determine which isotope in Figure 7.2 will be about 47% as radioactive after 33 years. Show your reasoning.

isotope: ……. [2]

(f) The release of Iodine-131, Cesium-134 and Cesium-137 into the air, ground and seawater were monitored closely by the Japanese authority because they readily decay by emitting beta-particles and gamma-rays. For example, Cesium-137 decays into Barium-137 (Ba) through beta-emission.

95% of Cesium-137 decays to a metastable excited nuclear energy state (Barium-137m) and is responsible for all emission of gamma-rays, while the other 5% directly decays to the ground state (Barium-137) of the nucleus. Figure 7.4 illustrates the decay scheme of Cesium-137.

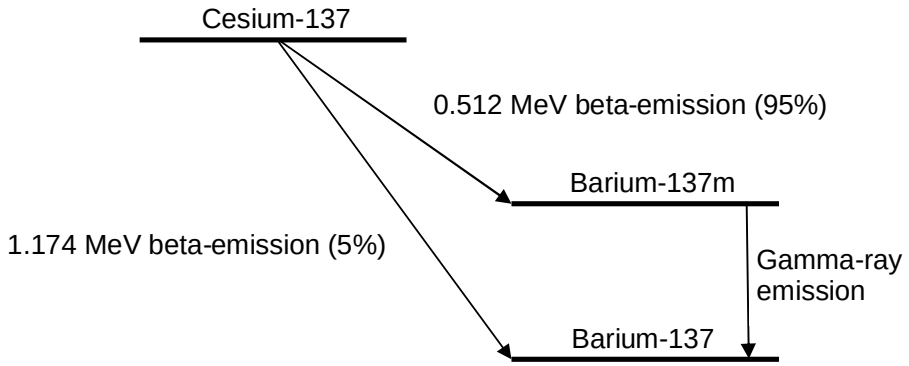
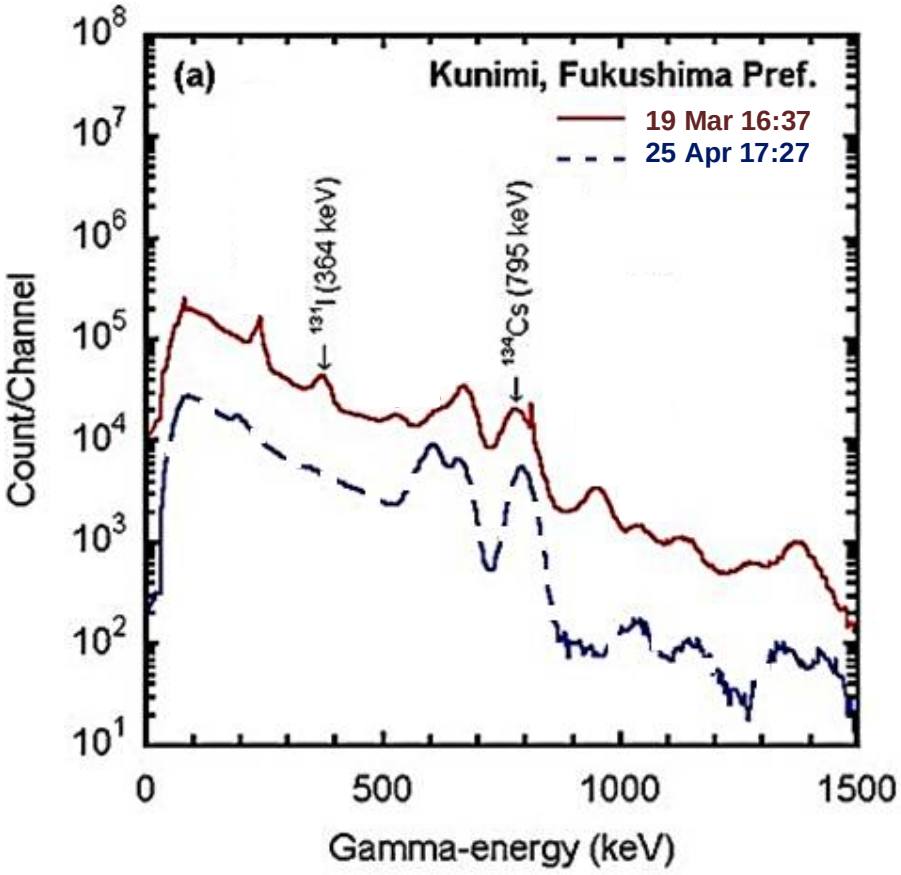


Figure 7.4

Determine the photon energy of the gamma-rays emitted during Cesium-137 decay.

energy = ……MeV [2]

(g) Gamma spectroscopy is a technique used by scientists to identify specific radioactive products from the many possible products that can be produced from the nuclear fission of Uranium-235. Figure 7.5 shows the gamma spectra measured at Kunimi, Fukushima Prefecture on both 19 March 2011 and 25 April 2011. Photon peaks generated by Tellurium-132, Iodine-131, -132, and Cesium-134, -136 and -137 were observed in the spectra given in Figure 7.5. The photon peaks corresponding to Iodine-131 and Cesium-134 are indicated in Figure 7.5.



Source: https://www.nature.com/articles/srep00087 (modified)

Figure 7.5

(i) Using your answer in **(f)**, draw an arrow on Figure 7.5 to show the position of Cesium-137 in the gamma spectra. Label this arrow Cs-137. [1]

(ii) Suggest why no photon peak of Iodine-131 was observed on 25 April 2011.

……

……

…… [1]

(h) Compare the three types of radioactive decay: alpha-decay, beta-decay and gamma-decay. Explain which of the three poses the greatest health risk if the source is

(i) outside the body,

……

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……

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…… [2]

(ii) ingested and entered the body.

……

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……

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…… [2]

[Total: 20]

END OF PAPER 2

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